A NEW METHOD OF ARRANGING VARIETY TRIALS INVOLVING A LARGE NUMBER OF VARIETIES

By F. YATES

(*Rothamsted Experimental Station, Harpenden, Herts*)

(With Two Text-figures)

I. Introduction

When a large number of varieties are to be compared in an agricultural field trial the experimenter is confronted with a difficult problem of arrangement. A Latin square containing the whole of the varieties is impossible, being both too unwieldy and requiring too many replications, and even randomised blocks containing the whole of the varieties are unsatisfactory, for the blocks are likely to contain too many plots to eliminate fertility differences efficiently.

A simple method of keeping the block size small is to select one or more varieties as controls and to divide the rest into sets, each set being arranged with the controls in a number of randomised blocks. Unfortunately this method has the disadvantage that comparisons between varieties in different sets are of lower accuracy than comparisons between varieties in the same set. There is the further disadvantage that a disproportionate number of plots is devoted to the control varieties, thus further lowering the efficiency.

The control method can, of course, also be used in conjunction with the Latin square method of design, each set of varieties together with the control or controls being arranged in the form of a Latin square.

Various systematic arrangements of control plots are also possible, the most widely used being that in which the control plot is repeated at fixed intervals along a line of plots. Most frequently in such arrangements a systematic arrangement of the new varieties is also adopted. This is wholly to be condemned, for not only is no valid estimate of error possible, but there is also the disadvantage that differences between neighbouring varieties are likely to be more accurately determined than those between widely separated varieties; moreover, there is a danger of serious disturbance from anything of the nature of a fertility wave. If, however, the new varieties are arranged at random all these objections are overcome, so far as the new varieties are concerned; only the comparisons with the controls are now liable to bias and lacking in an estimate of error.

Whatever the exact lay-out of the controls, any method using them suffers from the disadvantage that part of the land has to be devoted to them which would otherwise be devoted to experimental varieties, thus tending to lower the efficiency of the experiment.

To overcome this defect, and at the same time to avoid the use of excessively large blocks, the *pseudo-factorial* type of arrangement, described in this paper, is suggested. In this type of arrangement the varieties are divided into sets for comparison in more than one way, the sets of each division being so arranged that they cut across those of all the other divisions. Thus 100 varieties, numbered 00–99, may be divided into sets of 10 in two ways, the first group of 10 sets consisting of varieties 00–09, 10–19, 20–29, etc., and the second group of 10 sets consisting of varieties 00, 10, 20, ..., 90; 01, 11, 21, ..., 91; 02, 12, 22, ..., 92; etc. Each set of 10 can then be arranged in the field in the form of one or more randomised blocks of 10 plots each, or in the form of a 10×10 Latin square, according to the number of replications that are feasible.

Information on the difference of two varieties in the same set, such as varieties 00 and 01, will then accrue from two sources. Firstly there will be the direct comparison within the given set 00–09, and secondly 00 may be compared with the mean of 10, 20, ..., 90, and 01 may be compared with the mean of 11, 21, ..., 91 in the sets of the second group, the difference between the two means being determined from comparisons within the first group of sets. The difference between two varieties such as 00 and 11 not occurring in the same set may also be determined in various ways.

More elaborate arrangements of this type are clearly possible. Instead of two groups of sets three may be employed. With 125 varieties, for instance, three groups of 25 sets of 5 may be used, the first group of 25 sets being made up of varieties 1–5, 6–10, 11–15, ..., 121–125, the second group of varieties 1, 6, 11, 16, 21; 2, 7, 12, 17, 22; ..., 26, 31, 36, 41, 46; ..., and the third group of varieties 1, 26, 51, 76, 101; 2, 27, 52, 77, 102; .... With these three groups of sets, using 5×5 Latin squares, 15 replications would be required.

Moreover, the sets of the different groups need not necessarily be of the same size. Thus 90 varieties might be arranged in one group of 9 sets of 10, these being made up of varieties 1–10, 11–20, ..., 81–90, and one group of 10 sets of 9, made up of 1, 11, 21, ..., 81; 2, 12, 22, ..., 82; .... Similar arrangements are possible with three or more groups of sets.

The object of this paper is to discuss in detail the analysis and efficiency of these different types of pseudo-factorial arrangement. As a preliminary the efficiency of arrangements employing controls will be briefly considered.

II. Controls in conjunction with randomised blocks and Latin squares

We will first consider the use of controls in conjunction with ordinary randomised blocks or Latin squares.

Denote the number of plots per block by k, the number of controls per block by c, the number of varieties other than controls by v, the number of replicates of each variety other than controls by r, and the error variance per plot by σ^2.

The number of sets of varieties is then $v/(k-c)$, and the total number of plots N is $rkv/(k-c)$.

The error variances of the various types of comparison are as follows:

Comparison between	Error variance
Two varieties (not controls) in one set	$2\sigma^2/r$
Two varieties not in the same set	$\frac{2\sigma^2}{r}\left(1+\frac{1}{c}\right)$
Mean of controls and one other variety	$\frac{\sigma^2}{r}\left(1+\frac{1}{c}\right)$
One control and one other variety	$\frac{\sigma^2}{r}\left(1+\frac{1}{c}\right)+\frac{\sigma^2}{r}\frac{k-c}{v}\left(1-\frac{1}{c}\right)$

Since all the comparisons have not the same error variance some criterion for assessing the accuracy of the experiment as a whole will be required. The following suggest themselves:

(1) The greatest error variance of any comparison.

(2) The mean error variance of all comparisons.

(3) The mean of the information on all comparisons, *i.e.* the mean of the reciprocals of the error variances of all comparisons.

Criterion (3), which at first sight appears the logical one, does not, however, take into account the advantages that accrue from having all the comparisons of equal accuracy, and for some purposes (1) might be considered the best single criterion. Criterion (2) has the advantage that it has in general a simpler algebraic expression than (3). It will be somewhat greater than (3), though only very slightly so when the variation in accuracy is only small. In what follows, therefore, we shall limit ourselves to the first two criteria, basing our general conclusions on the second.

To determine the optimum number of controls with a given number of plots per block we shall require to determine the minimum of whatever criterion we consider appropriate. We shall first take as a criterion the greatest error variance of any comparison, namely

$$\frac{2\sigma^2}{r}\left(1+\frac{1}{c}\right).$$

This is equal to

$$\frac{2\sigma^2 vk\,(c+1)}{N\,(k-c)\,c}.$$

In the case in which the controls are standard varieties we require a value of c which makes

$$\text{const.}\times\frac{(c+1)}{(k-c)\,c}$$

a minimum. This is given by

$$c\,(k-c)+(c+1)\,(2c-k)=0,$$

$$c=-1+\sqrt{1+k}.$$

The mean error variance of all comparisons, including controls, is

$$V_m=\frac{2\sigma^2 k}{N\,(v+c)\,(v+c-1)}\left[\frac{v^2}{c\,(k-c)}\{(v+c)\,(c+1)-k\}+(v+c)\,(c-1)\right].$$

If the controls are themselves new varieties, and if the mean error variance of all comparisons is chosen as the criterion, the equations for c are more complicated, but will not lead to any widely different values when the number of varieties is large, for comparisons between varieties not in the same set then predominate. In practice only integral values of c are possible, and after calculating c from the above equation the relative mean error variances may be calculated for the two nearest integral values of c in order to decide on which value to adopt.

The efficiency of the arrangement clearly not only depends on the number of controls per block but also on the choice of block size k, hitherto taken as fixed. The error variance per plot will be some function of the block size, and until this function is known it is not possible to determine the most efficient block size. In general, however, it will be found that so far as arrangements in randomised blocks are concerned the method is unlikely to be much more efficient than randomised blocks including all varieties, and on uniform land may be much less efficient. Since it is always likely to be less efficient than the use of systematic controls and pseudo-factorial arrangements there is no need to investigate the matter in more detail here. It need only be noted that the use of randomly placed controls may make possible the employment of Latin squares where these would otherwise be impossible, and for this reason may possess a certain limited usefulness.

In the experiments at Rothamsted and its associated centres, for example, it was found that if no restrictions had been imposed the error variance of the 5×5 Latin squares carried out between 1927 and 1934 would have been increased on the average in the ratio 2·49 : 1. If we were conducting an experiment on 25 varieties we might take one as a control and divide the rest into six sets of four, using six 5×5 Latin squares in all. We should then have $V_m = 0{\cdot}72\sigma^2$. The corresponding variance for randomised blocks of 25 plots would be $V_B = \frac{1}{3}\sigma'^2$. This gives

$$V_B : V_m = 0{\cdot}462\sigma'^2/\sigma^2 = 1{\cdot}15.$$

Thus the method of controls is 15 per cent. more efficient than that of randomised blocks.

There is nothing in the pseudo-factorial type of arrangement which prevents the use of Latin squares instead of randomised blocks, but a greater number of replicates will be required, and this may be impracticable. In the above example a 5×5 pseudo-factorial arrangement would require ten 5×5 Latin squares. Should this number of replicates be possible, however, the pseudo-factorial arrangement (as will be shown later in the paper) would give an efficiency of $\frac{4}{6} \times 2{\cdot}49 = 1{\cdot}66$, and is thus a much better arrangement than the method of controls.

III. Systematic arrangement of controls

One of the disadvantages of the use of controls in conjunction with ordinary randomised blocks is that each control plot is used to furnish information only on the fertility of the block in which it occurs, though it may, for example, fall at the edge of that block and be in fact just as good an indicator of the fertility of plots in its neighbourhood belonging to the next block.

Many methods of utilising the information of systematically arranged control plots have been advocated from time to time, some of them very complicated, and most of them open to serious statistical objections. If, however, the control plots are regarded merely as indicators of the fertility in their neighbourhood and a proper randomisation process is applied to the varieties allotted to all the remaining plots, a statistically valid arrangement for the comparison of the varieties other than controls will result, and appropriate methods of analysis are available.

The simplest type of arrangement of this nature is that in which the plots are arranged in one or more lines, the controls being placed at equal intervals along the lines. Thus, for example, 12 new varieties might be compared in three randomised blocks with controls at every 5th plot. The following arrangement is of this type:

Block I

$C_1 \quad a \quad l \quad b \quad k \quad C_2 \quad g \quad e \quad f \quad h \quad C_3 \quad c \quad d \quad j \quad i \quad C_4$

Block II

$C_5 \quad j \quad g \quad l \quad k \quad C_6 \quad c \quad e \quad d \quad b \quad C_7 \quad a \quad f \quad i \quad h \quad C_8$

Block III

$C_9 \quad k \quad c \quad g \quad a \quad C_{10} \quad i \quad l \quad j \quad b \quad C_{11} \quad f \quad e \quad k \quad d \quad C_{12}$

In analysing such an experiment the experimenter is at liberty to use any function of the controls that he fancies as a measure of the fertility of the neighbouring plots. For practical reasons he will be well advised to choose a simple function, for the advantages accruing from the more complicated functions that have sometimes been suggested are likely to be more theoretical than real.

The simplest function is the mean of the two control plots between which the plot under consideration is situated, but the weighted mean of the two controls, with weights inversely proportional to their distances, would appear to be more suitable. Thus the appropriate values for the first four experimental plots in the above arrangement (varieties a, l, b, k respectively) would be

$$\tfrac{4}{5}C_1+\tfrac{1}{5}C_2,\ \tfrac{3}{5}C_1+\tfrac{2}{5}C_2,\ \tfrac{2}{5}C_1+\tfrac{3}{5}C_2,\ \tfrac{1}{5}C_1+\tfrac{4}{5}C_2.$$

Having decided on the measure of fertility, the experimenter is equally at liberty to use this measure in whatever way appears desirable. He may take the differences between it and the yields of the experimental plots, or he may express the experimental yields as percentages of the corresponding fertility measures, but here again he will be well advised to adopt some simple procedure; in particular percentages are unlikely to possess any advantages over differences. When the corrected values have been obtained it is only necessary to analyse them in the same manner as the results of an ordinary randomised block experiment.

There is, however, a more effective way of using these fertility measures. Such measures are of the nature of concomitant observations, and can be treated in the same manner as other information of this type, such as the yields of a preliminary uniformity trial, namely, by the procedure of the analysis of covariance. The corrected values will then be of the form $y_u - bf_u$, where y_u is the uncorrected yield of the u'th plot, f_u the fertility measure, and b is a constant whose value is determined from the observations themselves, being such that the error of the corrected yields is minimised. By this procedure the danger of overcorrection for the fertility measures, which will almost certainly occur if straight differences are taken, will be avoided, and only that weight will be attached to the fertility measures which is justified by the results of the particular experiment under consideration.

In cases where the majority of the variation in plot yields is due to causes other than fertility differences, or where the control plots provide an inadequate representation of such differences, a value of b much smaller than unity may be anticipated, and the covariance method will then be markedly the more accurate. In the limiting case where there is no association between the yields of neighbouring plots, so that the plot errors may be regarded as independent, with, say, a variance σ^2, the variances of the fertility indices of each set of four plots (with controls at intervals of five plots) are 17/25, 13/25, 13/25 and 17/25 times σ^2 respectively, the mean being $\frac{3}{5}\sigma^2$. The covariance will in this case be zero, and therefore the variance of the differences $y_u - f_u$ will be $\frac{8}{5}\sigma^2$. (With controls at intervals of n plots this variance equals $(5n-1)\,\sigma^2/3n$.) The method of covariance will give b zero, and we shall therefore be using the unaltered yields, which will have a variance of σ^2. The loss of efficiency in this case by the use of the differences $y_u - f_u$ in place of $y_u - bf_u$ is therefore very considerable, ranging from $\frac{1}{3}$ when every alternate plot is a control to $\frac{2}{5}$ when the interval between controls is large.

Apart from the avoidable loss of information due to the use of differences $y_u - f_u$, or other arbitrary functions, in place of $y_u - bf_u$, allowance must also be made, when comparing the efficiency of systematic controls with ordinary randomised blocks, for the reduction in the number of plots available for the experimental varieties due to the existence of the controls. The ratio of the amounts of information tends in the limiting case just considered to the ratio of the number of non-control plots to the total number of plots.

There is, of course, nothing in the method of controls that precludes the use of a fertility index deduced from controls of more than one line of plots, and in certain cases, particularly with square plots, this may be advantageous. Thus a pattern such as the following might be adopted:

C_1	+	+	C_2	+	+	C_3	+	+	C_4
+	+	+	+	+	+	+	+	+	+
+	+	+	+	+	+	+	+	+	+
C_5	+	+	C_6	+	+	C_7	+	+	C_8

The weighted mean of the four nearest control plots could be taken as the fertility index for plots not in the same row or column as a control, that for the second plot of the second row, for example, being:

$$\tfrac{4}{9}C_1 + \tfrac{2}{9}C_2 + \tfrac{2}{9}C_5 + \tfrac{1}{9}C_6.$$

Clearly many such patterns are possible. Their relative merits can only be tested by extensive examination of uniformity trials on the type of land and with the type of crop on which it is desired to experiment.

IV. Pseudo-factorial arrangements in two equal groups of sets

For two equal groups of sets to be possible the number of varieties must be a perfect square, say p^2. The variety in the u'th set of the first group and the v'th set of the second will be denoted by the pair of numbers uv. Its mean yield over the replicates of the first group of sets will be denoted by x_{uv}, and its mean yield over the replicates of the second group of sets by y_{uv}.

For purposes of computation the mean yields of each group of sets should be set out in squares of the form of Table I.

Table I

	First group					Second group					
Set	1	2	...	p	Mean	Set					Mean
	x_{11}	x_{21}	...	x_{p1}	$\bar{x}_{.1}$	1	y_{11}	y_{21}	...	y_{p1}	$\bar{y}_{.1}$
	x_{12}	x_{22}	...	x_{p2}	$\bar{x}_{.2}$	2	y_{12}	y_{22}	...	y_{p2}	$\bar{y}_{.2}$
	...				...	...	...				...
	x_{1p}	x_{2p}	...	x_{pp}	$\bar{x}_{.p}$	p	y_{1p}	y_{2p}	...	y_{pp}	$\bar{y}_{.p}$
Mean	$\bar{x}_{1.}$	$\bar{x}_{2.}$	...	$\bar{x}_{p.}$	$\bar{x}_{..}$	Mean	$\bar{y}_{1.}$	$\bar{y}_{2.}$	...	$\bar{y}_{p.}$	$\bar{y}_{..}$

The marginal means, or totals, will be required as indicated. In the actual numerical computations the use of totals is more convenient, but the corresponding algebraic formulae can be more neatly expressed (and remembered) in terms of the means; the substitution of totals in place of means for numerical work will present no difficulty.

The equations for estimating the varietal differences can be derived directly by the classical method of least squares, fitting constants for sets and for varieties. This method will also conveniently provide expressions for the various components of the sum of squares in the analysis of variance. The errors of the various types of comparison are, however, better derived by methods recently developed in connection with the analysis of factorial experiments as described by Fisher (1) and Yates (3). We propose, therefore, to use this line of approach in the present paper.

An experimental arrangement in two equal groups of sets is analogous to a factorial experiment involving two factors (each with p values) in which the main effects of one factor are confounded in one half (the first group) of the replications and the main effects of the other factor are confounded in the other half (the second group). In such an experiment the p^2-1 treatment degrees of freedom are divisible into

First factor	$p-1$
Second factor	$p-1$
Interactions	$(p-1)^2$

The main effects of the first factor will be estimated from the second group of replications only, and those of the second factor from the first group only, the precision of each being one-half that of an unconfounded experiment with the same error variance per plot. The interactions will be unaffected by the confounding and will have estimates of full precision.

In the corresponding pseudo-factorial varietal trial the estimates required are not those of main effects and interactions but differences between single varieties. Such differences must therefore be expressed in terms of the main effects and interactions.

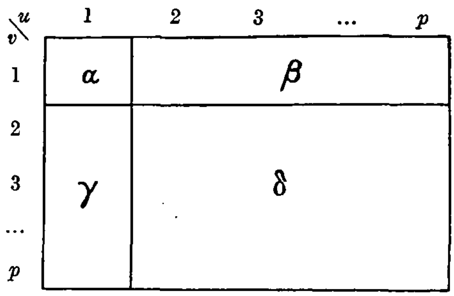

Fig. 1.

We will first consider the estimate of the deviation of any one variety from the mean of all the varieties. Let the true yields per plot of the varieties be represented by τ_{uv}, and take α, β, γ and δ to represent the sums of the τ's contained in the compartments of the two-way table represented in Fig. 1: so that $\alpha=\tau_{11}$, $\beta=\tau_{21}+\tau_{31}+\ldots\tau_{p1}$, etc. Further take

$$
\begin{aligned}
A &= (p-1)^2\alpha-(p-1)(\beta+\gamma)+\delta,\\
B &= (p-1)(\alpha+\beta)-(\gamma+\delta),\\
C &= (p-1)(\alpha+\gamma)-(\beta+\delta),\\
D &= \alpha+\beta+\gamma+\delta.
\end{aligned}
$$

The quantity A is a component of the interactions of the two factors and is therefore estimated from both groups of replications, *i.e.* from the mean of the x's and y's. B is a component of the main effects of the second factor and is therefore estimated from the first group of replications, *i.e.* from the x's only. Similarly C is estimated from the second group of replications, *i.e.* from the y's only.

Now $p^2\tau_{11}=A+B+C+D$, and replacing D by $p^2\bar{\tau}_{..}$ we have

$$p^2(\tau_{11}-\bar{\tau}_{..})=A+B+C,$$

and the estimate $t_{11}-\bar{t}_{..}$ of $\tau_{11}-\bar{\tau}_{..}$ is therefore given by

$$\begin{aligned} p^2(t_{11}-\bar{t}_{..}) = {} & (p-1)^2\tfrac{1}{2}(\alpha_x+\alpha_y)-(p-1)\tfrac{1}{2}(\beta_x+\beta_y+\gamma_x+\gamma_y)+\tfrac{1}{2}(\delta_x+\delta_y) \\ & +(p-1)(\alpha_x+\beta_x)-(\gamma_x+\delta_x) \\ & +(p-1)(\alpha_y+\gamma_y)-(\beta_y+\delta_y), \end{aligned}$$

which after some simplification, replacing $\bar{t}_{..}$ by $\frac{1}{2}(\bar{x}_{..}+\bar{y}_{..})$, reduces to the final generalised expression

$$t_{uv}=\tfrac{1}{2}(x_{uv}+y_{uv}-\bar{x}_{u.}+\bar{x}_{.v}+\bar{y}_{u.}-\bar{y}_{.v}).$$

Differences between the adjusted values t_{uv} will give efficient estimates of the varietal differences freed from block effects.

The variance of the difference of two such adjusted means can be determined in a similar manner by considering a 2×2 element of four values.

Consider the four values

$$\begin{matrix} \tau_{11} & \tau_{21} \\ \tau_{12} & \tau_{22} \end{matrix}$$

The differences between these may be represented as the main effects and interactions of this two-way table, say

$$\begin{aligned} U &= \tfrac{1}{2}\{(\tau_{21}+\tau_{22})-(\tau_{11}+\tau_{12})\}, \\ V &= \tfrac{1}{2}\{(\tau_{12}+\tau_{22})-(\tau_{11}+\tau_{21})\}, \\ \{U.V\} &= \tfrac{1}{2}\{(\tau_{11}+\tau_{22})-(\tau_{12}+\tau_{21})\}. \end{aligned}$$

Now take α', β', γ' and δ' to represent the sums of the τ's contained in the compartments of the two-way table represented in Fig. 2:

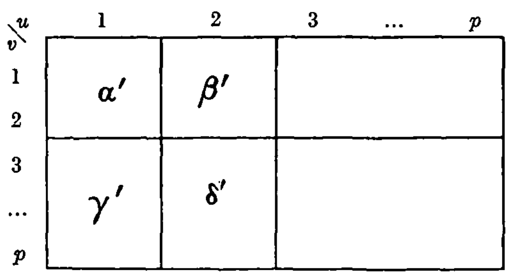

Fig. 2.

so that $\alpha' = \tau_{11} + \tau_{12}$, etc. Further take

$$E = (p-2)(\alpha' - \beta') - 2(\gamma' - \delta'),$$
$$F = \alpha' - \beta' + \gamma' - \delta'.$$

We have

$$-2pU = p(\alpha' - \beta') = E + 2F.$$

The quantity E is a component of interaction between the two factors and therefore its estimate has a variance

$$V(E) = \frac{\sigma^2}{r}\{(p-2)^2 4 + 8(p-2)\} = \frac{4\sigma^2}{r} p(p-2),$$

while F is a component of the main effect of the first factor and therefore its estimate has a variance

$$V(F) = \frac{2\sigma^2}{r}(2p) = \frac{4\sigma^2}{r} p.$$

Hence

$$V(U) = \frac{1}{4p^2}\{V(E) + 4V(F)\} = \frac{\sigma^2}{rp}(p+2) = V(V)$$

and since $\{U . V\}$ is a component of interaction its estimate has a variance of σ^2/r.

Now

$$\tau_{21} - \tau_{11} = U - \{U . V\}$$

and therefore the variance of the difference of two varieties having a set in common is

$$V(t_{21} - t_{11}) = \frac{\sigma^2}{rp}(p+2) + \frac{\sigma^2}{r} = \frac{2\sigma^2}{r} . \frac{p+1}{p},$$

since all the degrees of freedom concerned are orthogonal.

Similarly the variance of the difference of two varieties not having a set in common is

$$V(t_{22} - t_{11}) = V(U + V) = \frac{2\sigma^2}{r} . \frac{p+2}{p}.$$

It can easily be shown that the mean variance of all comparisons is equal to twice the expectation of the sum of squares of deviations between all adjusted varietal means divided by the associated number of degrees of freedom, $p^2 - 1$, thus providing a check. Resolving this sum of squares into its component parts we have the following expectations:

	Degrees of freedom	Expectation of mean square	Expectation of sum of squares
First factor	$p-1$	$2\sigma^2/r$	$2\sigma^2(p-1)/r$
Second factor	$p-1$	$2\sigma^2/r$	$2\sigma^2(p-1)/r$
Interactions	$(p-1)^2$	σ^2/r	$\sigma^2(p-1)^2/r$
Total	p^2-1		$\sigma^2(p+3)(p-1)/r$

Thus the mean variance of all comparisons is

$$V_m = \frac{2\sigma^2}{r} \cdot \frac{p+3}{p+1}.$$

Had there been no confounding the variance of every comparison would have been $2\sigma'^2/r$. The factor $(p+3)/(p+1)$ is therefore a measure of the increase in variance that results from the division of the varieties into sets when the error variance per plot is unaltered by the resultant reduction in block size. Its reciprocal, which may be called the *efficiency factor* of the arrangement, is a measure of the inherent strength of the arrangement. Values of this factor for various numbers of varieties are given in Table II, which also includes values of the similar factor for two groups of unequal sets described in section VII, and for the Latin square type of arrangement described in section VI, this latter being

$$(p+1)/(p+2\tfrac{1}{2}).$$

Table II. *Efficiency factors in two-dimensional arrangements*

No. of varieties Arrangement	25 5^2	30 5×6	36 6^2	42 6×7	49 7^2	64 8^2	100 10^2	144 12^2	256 16^2	400 20^2
Two groups of sets	0·75	0·764	0·778	0·788	0·8	0·818	0·846	0·867	0·895	0·913
Latin square grouping	0·8	—	0·824	—	0·842	0·857	0·880	0·897	0·919	0·933
Random controls	0·485	—	0·511	—	0·532	0·547	0·568	0·596	0·631	0·658

The table also shows the similar efficiency factors for the method of random controls described in section II. Since the error variance per plot is the same for all three types of arrangement the comparison of the three factors gives the relative efficiency of the three methods. It will be seen that the method of random controls is always very much less efficient than the corresponding pseudo-factorial arrangements.

The partition of the degrees of freedom in the analysis of variance is shown in Table III for the case of randomised blocks.

Table III

			Degrees of freedom	
Blocks	Between groups		1	$pr-1$
	Between sets	Group I	$p-1$	
		Group II	$p-1$	
	Within sets	Group I	$p(\frac{1}{2}r-1)$	
		Group II	$p(\frac{1}{2}r-1)$	
Varieties	First factor		$p-1$	p^2-1
	Second factor		$p-1$	
	Interactions		$(p-1)^2$	
Error	Between groups		$(p-1)^2$	$(p-1)(pr-p-1)$
	Within group I		$p(p-1)(\frac{1}{2}r-1)$	
	Within group II		$p(p-1)(\frac{1}{2}r-1)$	
		Total	p^2r-1	

The first factor component of the varietal sum of squares is computed from the values of group II only, and the second factor component from group I only. The varietal sums of squares are therefore equal to

First factor: $\frac{1}{2}r\,(pS\bar{y}_{u.}^2 - p^2 S\bar{y}_{..}^2)$.

Second factor: $\frac{1}{2}r\,(pS\bar{x}_{.v}^2 - p^2 S\bar{x}_{..}^2)$.

Interactions: $\frac{1}{4}r\,\{SS\,(x+y)^2 - pS\,(\bar{x}_{u.}+\bar{y}_{u.})^2 - pS\,(\bar{x}_{.v}+\bar{y}_{.v})^2 + p^2\,(\bar{x}_{..}+\bar{y}_{..})^2\}$.

The sum of squares for sets and groups is

$$\tfrac{1}{2}r\,\{p\,(S\bar{x}_{u.}^2 + S\bar{y}_{.v}^2) - \tfrac{1}{2}p^2\,(\bar{x}_{..}+\bar{y}_{..})^2\}.$$

If these four sums of squares are added together they total

$$\tfrac{1}{4}r\{SS(x+y)^2 + pS(\bar{x}_{u.}-\bar{y}_{u.})^2 + pS(\bar{x}_{.v}-\bar{y}_{.v})^2 - p^2(\bar{x}_{..}-\bar{y}_{..})^2 - p^2(\bar{x}_{..}+\bar{y}_{..})^2\}$$

which is the best form for computation, the sum of squares for varieties being obtained by subtraction of the sum of squares for sets and groups. The sum of squares for error between groups can likewise be obtained by subtraction of this total from the total sum of squares between all x and y, namely $\frac{1}{2}r\,\{SSx^2 + SSy^2 - p^2\,(\bar{x}_{..}+\bar{y}_{..})^2\}$.

V. Three-dimensional pseudo-factorial arrangements in three equal groups of sets

The number of varieties must be a perfect cube, say p^3. We shall only consider the type of arrangement in which each variety is specified by three co-ordinate numbers, u, v, w, vw denoting the set out of the p^2 of the first group in which the variety is located, uw and uv the similar sets of the second and third groups respectively. The mean yields of the variety in the $\frac{1}{3}r$ replicates of each of the three groups will be denoted by x_{uvw}, y_{uvw}, z_{uvw} respectively.

Each of these three groups of yields may be arranged in the form of a cube. The marginal means of each of these will be required and can be denoted by an obvious extension of the previous notation. The cube formed from the means, m_{uvw}, of the corresponding values of these three cubes will also be required.

By a similar process to that of the last section it will be found that the adjusted varietal means are given by

$$t_{uvw} = m_{uvw} + \tfrac{1}{2}\,(\bar{m}_{.vw} + \bar{m}_{u.w} + \bar{m}_{uv.}) - \tfrac{1}{2}\,(\bar{m}_{u..} + \bar{m}_{.v.} + \bar{m}_{..w}) - \tfrac{1}{2}\,(\bar{x}_{.vw} + \bar{y}_{u.w} + \bar{z}_{uv.}) + \tfrac{1}{2}\,(\bar{x}_{u..} + \bar{y}_{.v} + \bar{z}_{..w}),$$

which for computation is best written in the form

$$t_{uvw} = m_{uvw} + c_{.vw} + c_{u.w} + c_{uv.},$$

where $c_{.vw} = \frac{1}{2}\,(\bar{m}_{.vw} - \bar{x}_{.vw} - \bar{m}_{.v.} + \bar{y}_{.v.})$, etc.

Following the same procedure as in the previous section we may represent the differences of the cube of the eight τ's, $\tau_{111} \ldots \tau_{222}$, by the main effects U, V, W, and the interactions $\{U.V\}$, $\{U.W\}$, $\{V.W\}$, $\{U.V.W\}$ of the three-way table, each of these quantities being $\frac{1}{4}$ of the sum of four of the τ's less the sum of the other four.

It will be found that

$$V(U) = V(V) = V(W) = \frac{\sigma^2}{2rp^2}(p^2+2p+4),$$

$$V(\{U.V\}) = V(\{U.W\}) = V(\{V.W\}) = \frac{\sigma^2}{2rp}(p+1),$$

$$V(\{U.V.W\}) = \frac{\sigma^2}{2r}.$$

Since

$$\tau_{211}-\tau_{111} = U-\{U.V\}-\{U.W\}+\{U.V.W\},$$

$$\tau_{122}-\tau_{111} = V+W-\{U.V\}-\{U.W\},$$

$$\tau_{222}-\tau_{111} = U+V+W+\{U.V.W\},$$

we obtain

$$V(t_{211}-t_{111}) = \frac{2\sigma^2}{rp^2}(p^2+p+1),$$

$$V(t_{122}-t_{111}) = \frac{\sigma^2}{rp^2}(2p^2+3p+4),$$

$$V(t_{222}-t_{111}) = \frac{\sigma^2}{rp^2}(2p^2+3p+6).$$

The mean variance of all comparisons is found to be

$$V_m = \frac{\sigma^2}{r}\,\frac{2p^2+5p+11}{p^2+p+1}.$$

The efficiency of this type of arrangement is therefore

$$2(p^2+p+1)/(2p^2+5p+11).$$

The values of this factor for various numbers of varieties (including some involving unequal sets, as described in section VIII) are shown in Table IV, which also gives the corresponding factors for the method of controls for comparison.

Table IV. *Efficiency factors in three-dimensional arrangements*

No. of varieties Arrangement	64 4^3	80 $4^2 \times 5$	100 4×5^2	125 5^3	216 6^3	343 7^3	512 8^3
Factorial arrangement	0·667	0·684	0·702	0·721	0·761	0·792	0·816
Random controls	0·393	—	—	0·416	0·455	0·484	0·506

The efficiency factors of these three-dimensional arrangements are decidedly lower than those of the two-dimensional arrangements given in Table II. This, and the greater computational labour they entail, must be offset against the greater reduction in block size.

The partition of the degrees of freedom is similar to that already given for the classification into two groups of sets. The sum of squares due to treatments in the analysis of variance can be built up from its components in the same manner as before, but for practical purposes it is best calculated from the formula

$$rSSSt_{uvw}m_{uvw} - \tfrac{1}{3}rpSS\,(\bar{t}_{.vw}\,\bar{x}_{.vw} + \bar{t}_{u.w}\,\bar{y}_{u.w} + \bar{t}_{uv.}\bar{z}_{uv.}).$$

The sum of squares due to the component of error between groups can be calculated by deducting this, and the sums of squares due to sets, groups and mean, which three latter together total

$$\tfrac{1}{3}rpSS\,(\bar{x}_{.vw}{}^2 + \bar{y}_{u.w}{}^2 + \bar{z}_{uv.}{}^2),$$

from the total sum of squares arising from the values of the three cubes, namely,

$$\tfrac{1}{3}rSSS\,(x_{uvw}{}^2 + y_{uvw}{}^2 + z_{uvw}{}^2).$$

VI. Two-dimensional pseudo-factorial arrangements in three groups of sets forming a Latin square

If a number of varieties forming a perfect square is classified in two ways as in Table I, and in addition in a third way such that every set of the third group contains one and only one variety from each of the sets of the first group and also from each of the sets of the second group, so that the three classifications fulfil the conditions of a Latin square, each set of varieties in each of the three groupings may be compared by means of randomised blocks or a Latin square.

The analytical solution remains comparatively simple. Denote each of the p^2 varieties by three numbers u, v, w (indicating the sets of the three groups to which that variety belongs) and the mean yields for the three groupings by x_{uvw}, y_{uvw} and z_{uvw} respectively, their mean being m_{uvw}. It will be noted that any two of the three numbers are sufficient to specify the variety.

The estimates of the varietal differences are given by differences of the quantities

$$t_{uvw} = m_{uvw} + \tfrac{1}{2}\,(\bar{m}_{u..} + \bar{m}_{.v.} + \bar{m}_{..w}) - \tfrac{1}{2}\,(\bar{x}_{u..} + \bar{y}_{.v.} + \bar{z}_{..w}).$$

The variance of the differences of two varieties occurring in the same set of one of the three groupings is given by

$$V\,(t_{uvw} - t_{uv'w'}) = \frac{2\sigma^2}{r}\left(1 + \frac{1}{p}\right),$$

as in the classification into two groups of sets, and the variance of the difference of two varieties not occurring in the same set is given by

$$V\ (t_{uvw}-t_{u'v'w'})=\frac{2\sigma^2}{r}\left(1+\frac{3}{2p}\right).$$

The mean variance of all comparisons is

$$V_m=\frac{2\sigma^2}{r}\cdot\frac{p+2\frac{1}{2}}{p+1}.$$

The sum of squares for (mean + groups + sets + treatments) is equal to

$$r\left[SSm_{uvw}{}^2+\tfrac{1}{2}p\ \{S\ (\bar{x}_{u..}-\bar{m}_{u..})^2+S\ (\bar{y}_{.v.}-\bar{m}_{.v.})^2+S\ (\bar{z}_{..w}-\bar{m}_{..w})^2\}\right.$$
$$\left.-\frac{p^2}{6}\ \{(\bar{x}_{...}-\bar{m}_{...})^2+(\bar{y}_{...}-\bar{m}_{...})^2+(\bar{z}_{...}-\bar{m}_{...})^2\}\right].$$

This can be divided into its components in the same manner as the classifications into two groups of sets.

It will be noted that the variance of differences between treatments not in the same set is reduced by the comparisons of the third classification, but that the variance of differences between treatments in the same set remains the same, having as before the factor $\left(1+\frac{1}{p}\right)$. Now when the number of varieties is the square of a prime number (and also in certain other cases) further divisions in sets can be made such that each set contains one and only one variety of every set in every other group. (Each group of sets will constitute with each of the other groups a Graeco-Latin square.) If all the $p+1$ such groups of sets are included in the arrangement, every variety occurs with every other variety in one set and one only. Consequently all comparisons are of equal accuracy, and have in fact a variance of $\frac{2\sigma^2}{r}\left(1+\frac{1}{p}\right)$ equal to that of the most accurate comparison in the divisions into two and three groups of sets(4).

VII. Pseudo-factorial arrangements in two unequal groups of sets

So far we have considered only arrangements in which the number of varieties is the same in all sets. This necessitates that the number of varieties included in the experiment is a perfect square or a perfect cube. This limitation may be overcome if the number in each set is different for the different groups. In particular a two-factor classification may have q varieties in each of the p sets of the first group, and p varieties in each of the q sets of the second group, giving pq varieties in all, with u ranging from 1 to p and v from 1 to q.

An additional difficulty immediately presents itself in arrangements of this type. For since the sets in the two groups are of unequal size the blocks or Latin squares by which the comparisons within sets are made will also be of unequal size, and consequently the error variances per plot (σ_x^2 and σ_y^2 say) cannot be assumed equal for both groups.

Take the weights of the varietal means x_{uv} and y_{uv} of the two groups to be λ and μ respectively, so that $\lambda = r/2\sigma_x^2$ and $\mu = r/2\sigma_y^2$. The quantities A, B and C of section IV must be replaced by

$$\begin{aligned} A &= (p-1)(q-1)\alpha - (q-1)\beta - (p-1)\gamma + \delta, \\ B &= (q-1)(\alpha+\beta) - (\gamma+\delta), \\ C &= (p-1)(\alpha+\gamma) - (\beta+\delta). \end{aligned}$$

In estimates involving both x and y the weighted mean must be taken. Thus the quantity $\frac{1}{2}(\alpha_x + \alpha_y)$ must be replaced by $(\lambda\alpha_x + \mu\alpha_y)/(\lambda+\mu)$, etc. Estimates involving only x or y will remain unchanged.

This leads by the same procedure as before to the expression

$$(\lambda+\mu)\, t_{uv} = \lambda x_{uv} + \mu y_{uv} - \lambda(\bar{x}_{u.} - \bar{y}_{u.}) + \mu(\bar{x}_{.v} - \bar{y}_{.v}).$$

In order to obtain the variance of differences of the t's E must be redefined as

$$E = (q-2)(\alpha' - \beta') - 2(\gamma' - \delta'),$$

giving $-2qU = E + 2F$, and

$$V(E) = \frac{4q(q-2)}{\lambda+\mu}, \quad V(F) = \frac{2q}{\mu};$$

so that

$$\begin{aligned} V(U) &= \frac{1}{q}\left[\frac{2}{\mu} + \frac{q-2}{\lambda+\mu}\right], \\ V(V) &= \frac{1}{p}\left[\frac{2}{\lambda} + \frac{p-2}{\lambda+\mu}\right], \\ V\{U.V\} &= \frac{1}{\lambda+\mu}. \end{aligned}$$

Thus

$$\begin{aligned} V(t_{21} - t_{11}) &= \frac{2}{q}\left[\frac{1}{\mu} + \frac{q-1}{\lambda+\mu}\right], \\ V(t_{12} - t_{11}) &= \frac{2}{p}\left[\frac{1}{\lambda} + \frac{p-1}{\lambda+\mu}\right], \\ V(t_{22} - t_{11}) &= \frac{2}{pq}\left[\frac{q}{\lambda} + \frac{p}{\mu} + \frac{pq-p-q}{\lambda+\mu}\right], \end{aligned}$$

the mean variance of all comparisons being

$$V_m = \frac{2}{pq-1}\left[\frac{q-1}{\lambda} + \frac{p-1}{\mu} + \frac{(p-1)(q-1)}{\lambda+\mu}\right].$$

If the difference in weight is ignored and the t's estimated by the unweighted expression of section IV all components estimated from both x and y will be estimated from the unweighted means and will therefore have their variances increased in the ratio

$$\frac{1}{4}\left(\frac{1}{\lambda}+\frac{1}{\mu}\right):\frac{1}{\lambda+\mu}=\frac{(\lambda+\mu)^2}{4\lambda\mu}.$$

Consequently the variances of the differences of the unweighted t's may be immediately obtained by replacing the fraction $1/(\lambda+\mu)$ by $(\lambda+\mu)/4\lambda\mu$ wherever it occurs. The fractional increase of these components will therefore be

$$\frac{(\lambda+\mu)^2}{4\lambda\mu}-1=\frac{(\lambda-\mu)^2}{4\lambda\mu},$$

and the fractional increases in the total variances will therefore always be slightly less than this.

The percentage values of this fraction are

$\lambda:\mu$	2 : 1	3 : 2	4 : 3	5 : 4	6 : 5
Percentage	12·5	4·2	2·1	1·2	0·83

The loss of information by the use of unweighted means is therefore very small provided the error variances of the two groups are not widely different. This is usually the case in practice if p and q are not widely different. Nor is there ever any reason why they should be. With 90 varieties, for instance, a 9×10 arrangement would be chosen in preference to a 6×15 arrangement, since the nearer the arrangement approaches to symmetry the more efficient it will be. With 95 varieties a 10×10 arrangement with 5 additional varieties would undoubtedly be better than a 19×5 arrangement and probably better than a 12×8 arrangement with one additional variety.

When the weights are equal, so that $\lambda=\mu=r/2\sigma^2$, the variances of the varietal comparisons become

$$V\,(t_{21}-t_{11})=\frac{2\sigma^2}{r}\left(1+\frac{1}{q}\right),$$

$$V\,(t_{12}-t_{11})=\frac{2\sigma^2}{r}\left(1+\frac{1}{p}\right),$$

$$V\,(t_{22}-t_{11})=\frac{2\sigma^2}{r}\left(1+\frac{1}{p}+\frac{1}{q}\right),$$

$$V_m=\frac{2\sigma^2}{r}\,\frac{(p+1)\,(q+1)-4}{pq-1}.$$

If the weights are unequal these expressions may be used without appreciable error in place of the exact expressions in all cases in which it is admissible to ignore the differences of weights.

There is another cause of unequal error variance in the two groups. If these occupy different parts of the field it may be that the fertility differences of one part are less adequately eliminated by the blocks than those of the other part, or one part of the field may be more irregular than the other part. Consequently it is advisable to randomise the sets of treatments allotted to the different blocks (in so far as this is compatible with the arrangement of the blocks) whether p and q are unequal or not. This has been done in the examples which follow.

Finally, if the comparisons within sets are made by means of Latin squares instead of randomised blocks, the number of replicates in the two groups will be unequal, being p and q respectively, giving rise to a further inequality in weights, though this inequality is likely to be compensated to a certain extent by a lower error variance in the smaller squares. Even in this case, therefore, weighting will not always be necessary, though it would seem advisable to examine the two components of error within sets before making a decision.

VIII. Three-dimensional pseudo-factorial arrangements in three unequal groups of sets

Let the varieties be classified according to three numbers u, v, w, ranging from 1 to p, 1 to q and 1 to r respectively, the corresponding weights being λ, μ and ν.

Arguments similar to those developed in the last section apply here also. The exact expressions for t_{uvw} are given by

$$(\lambda+\mu+\nu)\, t_{uvw} = \lambda\,(x_{uvw}-\bar{x}_{.vw}) + \frac{\lambda\nu}{\lambda+\mu}(\bar{x}_{uv.}-\bar{x}_{.v.}) + \frac{\lambda\mu}{\lambda+\nu}(\bar{x}_{u.w}-\bar{x}_{..w})$$
$$+\bar{x}_{u..}\left(\frac{\mu\nu}{\lambda+\mu}+\frac{\mu\nu}{\lambda+\nu}\right) + \text{similar terms in } y \text{ and } z.$$

The variances of the different types of comparison are

$$V\,(t_{211}-t_{111}) = \frac{2}{qr}\left[\frac{1}{\lambda}+\frac{q-1}{\lambda+\mu}+\frac{r-1}{\lambda+\nu}+\frac{(q-1)\,(r-1)}{\lambda+\mu+\nu}\right],$$

$$V\,(t_{122}-t_{111}) = \frac{2}{pqr}\left[\frac{q}{\mu}+\frac{r}{\nu}+\frac{q(p-1)}{\lambda+\mu}+\frac{r(p-1)}{\lambda+\nu}\right.$$
$$\left.+\frac{qr-q-r}{\mu+\nu}+\frac{(qr-q-r)(p-1)}{\lambda+\mu+\nu}\right],$$

$$V\,(t_{222}-t_{111}) = \frac{2}{pqr}\left[\frac{p}{\lambda}+\frac{q}{\mu}+\frac{r}{\nu}+\frac{qr-q-r}{\mu+\nu}+\frac{pr-p-r}{\lambda+\nu}\right.$$
$$\left.+\frac{pq-p-q}{\lambda+\mu}+\frac{(p-1)\,(q-1)\,(r-1)+1}{\lambda+\mu+\nu}\right],$$

with similar expressions for $V\,(t_{121}-t_{111})$, etc.

If the differences in weights are ignored the fractional increase in components with divisor $\lambda+\mu$ will be $(\lambda-\mu)^2/4\lambda\mu$ as before, with similar increases with the divisors $\lambda+\nu$ and $\mu+\nu$. The fractional increase in the components with the divisor $\lambda+\mu+\nu$ will be

$$\frac{1}{9}\left(\frac{1}{\lambda}+\frac{1}{\mu}+\frac{1}{\nu}\right)(\lambda+\mu+\nu)-1=\frac{(\mu-\nu)^2}{9\mu\nu}+\frac{(\lambda-\nu)^2}{9\lambda\nu}+\frac{(\lambda-\mu)^2}{9\lambda\mu}.$$

The loss of information due to the use of unweighted means will therefore be of the same order as in the case of two unequal groups of sets. In practice p need not differ by more than one unit from q and r, which can be equal. This makes available a useful set of values for the total number of varieties which is not effectively increased by the inclusion of a greatest difference of 2 units between p, q and r. The possible numbers between 4^3 and 8^3 are:

64, 80, 100, 125, 150, 180, 216, 252, 294, 343, 392, 448, 512.

When the weights are equal the variances of varietal comparisons are:

$$V(t_{211}-t_{111})=\frac{\sigma^2/r'}{qr}(2qr+q+r+2),$$

$$V(t_{122}-t_{111})=\frac{\sigma^2/r'}{pqr}(2pqr+pq+pr+qr+2q+2r),$$

$$V(t_{222}-t_{111})=\frac{\sigma^2/r'}{pqr}(2pqr+pq+pr+qr+2p+2q+2r).$$

$$V_m=\frac{\sigma^2/r'}{pqr-1}\{2(p-1)(q-1)(r-1)+3(q-1)(r-1)+3(p-1)(r-1)+3(p-1)(q-1)+6(p-1)+6(q-1)+6(r-1)\},$$

r' being the number of replications.

IX. Numerical examples

In order to illustrate the various types of arrangement that are discussed in this paper we will take as an example the preliminary results of an experimental orchard of orange trees at the University of California Citrus Experiment Station at Riverside. The orchard was laid out for fertiliser trials, but during the first ten years of its existence was run as a uniformity trial without treatment.

The trial is fully reported and the results discussed by Parker and Batchelor (2). There were 199 plots in all, each of which contained eight trees in a single row, separated by a row of non-experimental oranges and grape fruit from the neighbouring plots. Each tree occupied a square of area 0·011 acre. Every attempt was made to select uniform land and to eliminate variability due to rootstocks and scions.

Table V. *Mean annual yields in pounds per tree*

	M	L	K	J	I	H	G	F	E	D
2	87*	103*	96*	103*	113*				100*	121*
4	95	106	119	104	120				127	140
6	111	110	109	99	140				116	141
8	79	112	99	115	129				139	137
10	98	113	116	110	122				130	128
12	102*	109*	122*	128*	113*				140	133*
14	93	126	137	128	108	132*			133*	138
16	100	116	129	119	113	130			100	107
18	93	115	112	119	120	132			124	105
20	98	116	127	135	125	131			116	105
22	98*	138*	132*	137	138*	116	120*		122	124*
24	103	†	121	119*	115	114*	123		118*	121
26	97	121	104	110	117	115	103			130
28	119	119	129	132	129	138	141			116
30	108	131	148	130	152	153	140			136
32	127	127*	135	134	126	129	133*	152*		139
34	119*	†	140*	134*	121*	139*	135	128		128*
36	135	147	146	147	150	143	127	138		136
38	120	137	119	124	136	142	148	129		133
40	121	138	135	139	142	150	133	131		122
42	114	128	157	146	144	142*	155	153*		138
44	107*	126*	143*	135*	126*	†	144*	165		144*
46						126	155	155		146
48						129	140	140		117
50						134	142	131		115
52						114	150	149*		122
54						114*	136*	†		108*

* Used as controls. † Yield missing.

Table V shows the mean yields of each plot from 1922 to 1927. The arrangement of the values in the table corresponds to the arrangement of the plots on the ground. The values of four of the plots were discarded for various reasons.

Table VI. *Residual mean squares for blocks of various sizes*

No. of plots per block	No. of blocks	Residual mean square	Plots discarded
199	1	247·43	—
64	3	219·15	L 24†, L 34†, H 44†, F 54†, D 2, D 4, D 6
49	4	192·16	F 54†, E 22, E 24
8	24	119·51	As for blocks of 64
7	28	121·65	As for blocks of 49
4	48	100·89	As for blocks of 64
2	96	104·70	As for blocks of 64

† Yield missing.

The residual mean squares after eliminating block differences are shown for blocks of various sizes in Table VI. These particular block sizes are those required for the examples we shall consider.

The residual mean square is reduced consistently by decrease in block size, except for blocks of two plots and of seven plots. About $2\frac{1}{2}$ times as much information is available on comparisons within blocks of four plots as is available on the average of all comparisons.

The mean square for blocks of two plots is not significantly larger than the mean square for blocks of four plots, but the reversal of the tendency to decrease is interesting, and may possibly indicate the existence of competition effects.

The following arrangements will be considered in detail:

(*a*) The comparison of 49 varieties using systematic controls.

(*b*) The comparison of 49 varieties in a 7×7 pseudo-factorial arrangement.

(*c*) The comparison of 64 varieties in a $4 \times 4 \times 4$ pseudo-factorial arrangement.

The relative accuracy of these and other possible arrangements will also be discussed.

(*a*) *Systematic controls*

In order to illustrate the method of systematic controls these have been taken at intervals of five or six plots. The actual plots selected are marked with an asterisk in Table V. The choice was made without reference to the yields, and where necessary by random selection. Missing plots were omitted, though it may be noted that by ill chance all four missing plots would have been chosen as controls.

The weighted means of the two neighbouring control plots may now be calculated for each of the experimental plots. Those of M 4–10, for instance, are 90, 93, 96, 99, and those of H 44–52 are [137·33], 132·67, 128, 123·33, 118·67, that in square brackets corresponding to the missing plot H 44. These means will be taken as the fertility indices of the plots. Fractions may be avoided by the use of multiples in the actual analyses.

The analyses of variance and covariance of the fertility indices and the corresponding actual yields are shown in Table VII, which also shows the analyses of variance for the differences of the indices and the yields and for the quantities $y - bf$. The value of b is given by

$$14817{\cdot}69/19885{\cdot}13 = 0{\cdot}7452.$$

The residual mean square for the differences equals 149·78, and that for the quantities $y-bf$ equals 141·90, the degrees of freedom being reduced by 1 in the second case.

In this example the value of b is not very far removed from unity; the difference between the two errors is consequently not very large, and the covariance method gives only 5 per cent. more information.

Table VII. *Analyses of variance and covariance of fertility indices* (f) *and plot yields* (y)

		Sums of squares and products				
		f^2	fy	y^2	$(y-f)^2$	$(y-bf)^2$
Blocks	2	5271·21	4768·10	4773·04	508·05	593·95
Residual	146	19885·13	14817·69	31617·63	21867·38	20576·02
Total	148	25156·34	19585·79	36390·67	22375·43	21169·97

To make allowance for the loss of information due to the reduction in the number of available plots these error variances must be multiplied by 195/149 (*i.e.* total number of plots/number of non-control plots) before comparison with that obtained in ordinary randomised blocks of 49 plots. The covariance method thus gives a value of 185·70 as compared with 192·16 for randomised blocks of 49 plots.

(*b*) 7 × 7 *pseudo-factorial arrangement*

The plots were divided without reference to the yields into 28 blocks of seven plots each. The particular set of varieties assigned to each block was chosen at random from all the sets for the reasons indicated at the end of section VII. The varieties within each set were then assigned to the plots of the chosen block at random.

Table VIII indicates the resulting arrangement, and should be read in conjunction with Table V. Three of the plots with missing yields have been included in the arrangement, as would be the case in practice.

When some of the yields are missing the correct way of conducting the analysis is to represent these missing yields by symbols, form the error mean square in the ordinary way, and determine the values of the missing yields which make this error mean square a minimum. The analysis can then be completed using these values.

There is no great difficulty in this procedure, but a simpler one is available which will usually give satisfactory results. This consists of giving a value to each missing yield which minimises the part of the error mean square contributed by all the replicates of the set for which the yield is missing. In the example before us there are only two replicates of each set, and this will therefore be effected when the difference between the value of the missing yield and that of the other replicate of the same variety is equal to the mean difference between the two replicates of all other varieties of these two sets.

Table VIII. *Arrangement of blocks and varieties: 7 × 7 pseudo-factorial arrangement*

	M	L	K	J	I	H	G	F	E	D
2	77	75	74	73	72				71	76
4	52	73	72	57	44				73	43
6	72	71	42	51	43				63	33
8	32	77	12	55	47				56	23
10	62	75	22	53	45				46	13
12	12	72	62	52	42				26	53
14	22	74	32	54	41	26			16	63
16	42	76	52	56	46	16			76	67
18	23	23	75	71	36	46			36	61
20	13	21	25	31	34	56			66	65
22	73	26	45	61	32	66	55		*	66
24	53	22†	65	11	37	76	51		*	64
26	43	24	15	21	33	36	56			62
28	33	27	35	41	35	11	52			34
30	63	25	55	51	31	17	57			54
32	26	45	14	77	57	15	54	17		64
34	22	46†	64	37	37	13	53	13		74
36	27	42	74	47	47	14	34	14		44
38	21	47	24	67	27	12	32	12		14
40	25	43	34	27	67	16	36	11		24
42	24	44	44	57	17	51	31	16		55
44	23	41	54	17	77	71†	35	15		65
46						61	33	17		75
48						31	37	11		15
50						21	66	13		45
52						41	65	12		35
54						11	64	*†		25

* Discarded. † Yield missing.

Blocks of 49 plots: (1) M, L, K 2–10; (2) K 12–44, J, I 2–20; (3) I 22–44, H, G 22–52; (4) G 54, F 32–52, E, D.

Here the yield of plot 24 of column L is missing. This plot corresponds to variety 22 in the first grouping, and the mean of all the other yields in this block is 123·3. The other replicate of 22 in the first grouping is the plot 34 of column M with yield 119, the mean of all the other plots in the block being 120·7. The difference is 123·3 − 120·7 = 2·6 and the required yield is therefore 119 + 2·6 = 122 say. The values of plot 34 of column L and plot 44 of column H are similarly found to be 125 and 118 respectively.

This procedure will slightly inflate the mean squares for treatments and for error between groups, their average value (when treatments produce no effect) and the average value of the mean square for error within groups being approximately in the ratio of 99 : 96. It will, moreover, sacrifice some of the available information on the varieties with missing plots. In this respect it will be roughly equivalent to the process of estimating a quantity ζ from three measures z_1, z_2 and z_3 of equal

Table IX. 7×7 *arrangement: sums of yields (less* 100) *of pairs of replicates in same grouping*

First group ($2x_{uv}$)

Set	1	2	3	4	5	6	7	Total
	69	36	107	34	22	45	10	323
	71	41*	86	60	69	79	22	428
	...	...	...	...	...	...	...	...
Total	590	305	498	373	298	433	115	2612

Second group ($2y_{uv}$)

Set								Total
1	33	44	64	46	72	63	37*	359
2	1	9	16	9	24	20	30	109
	...	...	...	...	...	...	...	...
Total	291	306	336	391	439	349	298	2410

* Including estimate of missing value.

Sum of both groups

$v \backslash u$	1	2	3	4	5	6	7	Total	$\frac{1}{2}(\bar{x}_{.v}-\bar{y}_{.v})$
1	102	80	171	80	94	108	47	682	− 1·3
2	72	50	102	69	93	99	52	537	+11·4
	...	...	...	...	...	...	...	...	...
Total	881	611	834	764	737	782	413	5022	+ 7·2
$\frac{1}{2}(\bar{y}_{u.}-\bar{x}_{u.})$	−10·7	0·0	−5·8	+0·6	+5·0	−3·0	+6·5	−7·2	

Adjusted values of treatment means (t_{uv})

$v \backslash u$	1	2	3	4	5	6	7
1	113·5	118·7	135·6	119·3	127·2	122·7	117·0
2	118·7	123·9	131·1	129·2	139·6	133·2	130·9
	...	...	...	...	...	...	...

accuracy by the expression $\frac{1}{2}\{\frac{1}{2}(z_1+z_2)+z_3\}$, instead of the mean $\frac{1}{3}(z_1+z_2+z_3)$. The variances of these two expressions being in the ratio $\frac{3}{8} : \frac{1}{3}$ or $\frac{9}{8}$, the amount of information lost will be $\frac{1}{9}$, which is not likely to be serious unless the variety involved turns out to be of special interest. If there are six replications, instead of four, the corresponding ratio is $\frac{5}{24} : \frac{1}{5}$ or $\frac{25}{24}$.

Table IX shows the sum of the two replicates of each variety for each grouping, arranged in the same manner as Table I. In order to save space only the values for varieties having $v=1$ or 2 are included. The sum of these two tables is also shown, together with the quantities $\frac{1}{2}(\bar{y}_{u.}-\bar{x}_{u.})$ and $\frac{1}{2}(\bar{x}_{.v}-\bar{y}_{.v})$, the first of the former of these quantities, $-10{\cdot}7$, for example, being equal to $\frac{1}{28}(291-590)$. From these values the adjusted values of the varietal means are derived. These are shown in the last line of Table IX, that for variety 11, for example, being given by

$$\tfrac{1}{4}.102-10{\cdot}7-1{\cdot}3+100=113{\cdot}5.$$

The analysis of variance is shown in Table X. The sum of squares for (mean + groups + sets + varieties) is equal to

$$142{,}788+6299{\cdot}86+5370{\cdot}71-208{\cdot}18=154{,}250{\cdot}39,$$

the terms being given in the same order as in the formula of section IV. The sum of squares for (mean + groups + sets) is 147,149·71. The difference gives the sum of squares for varieties, and the difference of the first quantity from the total sum of squares arising from all x_{uv} and y_{uv}, namely 158,077, gives the error between groups.

Table X. *Analysis of variance: 7 × 7 arrangement*

	Degrees of freedom	Sum of squares	Mean square	
Blocks (including groups and sets)	27	27928·63	1034·39	
Varieties	48	7100·68	147·93	
Error: Between groups	36	3826·61	106·29	121·64
Within groups	81*	9144·15	112·89	
Total	192*	48000·07	250·00	

* 3 deducted for missing yields.

The estimates of error between and within groups are about equal, as they should be. The combined error mean square, 110·86, may be compared with the treatment mean square, 147·93, by means of the z test. The value of z obtained is 0·1440, whereas the 5 per cent. point for these numbers of degrees of freedom is equal to 0·1913. The observed value (which is slightly inflated because of the method of estimating the missing plots) is thus well within the range that may be expected by chance.

The standard errors of the differences of the adjusted varietal means are as follows:

Two varieties in same set (row or column):

$$\sqrt{\frac{2\times 110{\cdot}86}{4}\left(1+\tfrac{1}{7}\right)}=\pm 7{\cdot}96.$$

Two varieties not in same set:

$$\sqrt{\frac{2\times 110{\cdot}86}{4}\left(1+\tfrac{2}{7}\right)}=\pm 8{\cdot}44.$$

If one of the varieties has a missing yield the standard errors will be increased approximately in the square root of the ratio $(\frac{1}{4}+\frac{3}{8}):(\frac{1}{4}+\frac{1}{4})$ or $\sqrt{5/4}$, and if both the varieties have missing yields by $\sqrt{3/2}$.

In order to compare the accuracy of this pseudo-factorial arrangement with an arrangement in randomised blocks the pooled estimate of error from varieties and error should be taken. This is 121·64. The corresponding error for randomised blocks of 49 plots is 192·16. The ratio $192{\cdot}16:121{\cdot}64=1{\cdot}580$, multiplied by the factor $(p+1)/(p+3)$ or 8/10, gives 1·264, so that the pseudo-factorial arrangement is 26 per cent. more efficient. If a 7×7 Latin-square pseudo-factorial arrangement had been adopted (this would require a multiple of 3 replications) the factor would be $(p+1)/(p+2\frac{1}{2})$ or 16/19, giving the ratio 1·331, *i.e.* a gain of a further 7 per cent.

Table XI. *Arrangement of blocks and varieties:* $4\times 4\times 4$ *pseudo-factorial arrangement*

	M	L	K	J	I	H	G	F	E	D
2	342	421	424	234	134				142	*
4	242	423	422	334	434				144	*
6	142	341	442	231	223				143	*
8	442	311	432	431	224				141	211
10	213	321	422	331	221				243	213
12	313	331	412	131	222				143	214
14	413	112	222	124	143	233			343	212
16	113	132	232	114	133	243			443	432
18	323	122	242	144	113	213			111	132
20	333	142	212	134	123	223			113	232
22	343	244	441	114	131	414	342		114	332
24	313	*†	421	214	141	411	341		112	222
26	431	224	431	414	121	412	344			122
28	434	214	411	314	111	413	343			322
30	432	234	424	444	321	121	313			422
32	433	241	444	443	323	221	312	433		322
34	323	*†	434	441	324	421	314	423		312
36	423	231	414	442	322	321	311	413		332
38	223	211	244	324	433	121	224	443		342
40	123	221	242	344	233	124	424	334		232
42	412	312	241	334	133	122	124	333		231
44	212	112	243	314	333	*†	324	332		233
46						123	132	331		234
48						141	134	411		144
50						341	133	111		444
52						241	131	211		344
54						441	311	*†		244

* Discarded. † Yield missing.

Blocks of 64 plots: (1) M, L, K; (2) J, I, H; (3) G, F, E, D.

(c) $4 \times 4 \times 4$ pseudo-factorial arrangement

For this arrangement the plots were divided without reference to the yields into 48 blocks of four plots each, and the 64 varieties assigned at random as in the 7×7 arrangement. Missing plots were, however, excluded. The actual arrangement obtained is shown in Table XI.

The first step in the computations is to set out the yields in order as in Table XII, where the yields of the first group and the sum of all the groupings are shown for varieties having $v=1$ or 2, together with their marginal totals. The yields of the other two groupings and of varieties having $v=3$ or 4, have been omitted to save space.

The next step is to prepare a table of the quantities $c_{.vw}$, $c_{u.w}$ and $c_{uv.}$. These are also shown in Table XII. Thus

$$-3{\cdot}38 = \tfrac{1}{24}\,365 - \tfrac{1}{8}\,156 - \tfrac{1}{96}\,1178 + \tfrac{1}{32}\,322.$$

Finally a table of the adjusted yields may be prepared by adding these corrections to one-third of the sum of all the groupings. Thus

$$132{\cdot}16 = 100 + \tfrac{1}{3}(84) - 3{\cdot}38 + 6{\cdot}03 + 1{\cdot}51.$$

The analysis of variance is shown in Table XIII. The sum of squares due to treatments is given by the sum of the products of the adjusted values and the variety totals of Table XII, less the sum of the products of the block totals and the corresponding marginal means of the adjusted values, $\bar{t}_{.vw}$, etc. (not shown). Omitting the 100's from the t's the first sum of products is 130,560·81, while the second is given by the sum of $27{\cdot}98 \times 156$, etc., and two similar sums for y and z, which three together total 124,189·42. The computation of the other sums of squares requires no comment.

The standard error of the difference of two varieties occurring in the same set (*i.e.* having two of the three numbers u, v, w the same) is

$$\sqrt{\frac{2 \times 100{\cdot}69 \times 21}{48}} = \pm 9{\cdot}39,$$

that of two varieties having one of the three numbers u, v, w the same is

$$\sqrt{\frac{100{\cdot}69 \times 48}{48}} = \pm 10{\cdot}03,$$

and that of two varieties having no common number is

$$\sqrt{\frac{100{\cdot}69 \times 50}{48}} = \pm 10{\cdot}24.$$

To compare the accuracy of the $4 \times 4 \times 4$ arrangement with an arrangement in ordinary randomised blocks of 64 plots the pooled residual

Table XII. $4 \times 4 \times 4$ *arrangement*

w	1					2					3					4					Sum				
u/v	1	2	3	4	$.vw$	1	2	3	4	$.vw$	1	2	3	4	$.vw$	1	2	3	4	$.vw$	$uv.$				$.v.$
											Yields of first group (x_{uvw})														
1	31	49	36	40	156	26	7	28	14	75	0	−2	2	−7	−7	37	19	32	10	98	94	73	98	57	322
2	53	29	43	39	164	30	21	16	36	103	21	20	19	35	95	55	48	44	33	180	159	118	122	143	542
	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	141	91	123	108	463	72	28	55	36	191	105	90	80	64	339	122	78	102	78	380	440	287	360	286	1373
	$u.w$				$..w$	$u.w$				$..w$	$u.w$				$..w$	$u.w$				$..w$	$u..$				...
											Sum of three groups ($3m_{uvw}$)														
1	84	123	75	83	365	70	72	89	51	282	36	58	45	69	208	78	71	102	72	323	268	324	311	275	1178
2	112	89	108	63	372	87	71	105	71	334	72	91	38	69	270	133	98	89	77	397	404	349	340	280	1373
	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	395	394	324	242	1355	260	241	359	171	1031	271	370	232	316	1189	362	314	336	308	1320	1288	1319	1251	1037	4895
											Corrections ($c_{.vw}$, $c_{u.w}$, $c_{uv.}$)														
					−3·38					+3·29					+10·45					+2·12	+1·51	−8·27	−5·70	−0·78	
					−6·98					−0·94					−2·61					−7·94	−2·83	−3·23	−6·24	+6·80	
					...					...					...					...	...	...	...	...	
	+6·03	−0·26	+9·95	+1·28		+5·97	+1·18	+2·22	+5·64		+6·91	+3·66	+14·53	−1·34		+1·74	−1·26	−4·72	−9·01						

Adjusted treatment means (t_{uvw})

132·16	129·09	125·87	124·79	134·10	120·20	129·48	125·15	130·87	125·17	134·28	131·33	131·37	116·26	125·70	116·33
133·55	119·20	132·73	122·10	131·20	120·68	130·04	135·17	125·47	128·15	118·35	125·85	135·30	120·24	110·77	115·52
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

variance of varieties and error, 100·89, must be multiplied by the factor $\frac{63}{42}$, giving 151·34. The ratio of the residual variance of blocks of 64 plots, 219·15, to this is 1·448, so that the pseudo-factorial arrangement is 45 per cent. more efficient.

Table XIII. *Analysis of variance:* $4 \times 4 \times 4$ *arrangement*

	Degrees of freedom	Sum of squares	Mean square	
Blocks (groups and sets)	47	33004·74	702·23	
Varieties	63	6371·39	101·13	100·89
Error between groups	81	8155·86	100·69	
Total	191	47531·99	248·86	

If instead of a $4 \times 4 \times 4$ arrangement an 8×8 arrangement had been adopted (an even number of replications would be required for this), the residual variance of blocks of eight plots, 119·51, would have to be multiplied by the factor $\frac{11}{9}$, giving a ratio of 1·500. With an 8×8 Latin-square pseudo-factorial arrangement the factor would be $\frac{21}{18}$, giving a ratio of 1·572.

X. Comparison of the relative efficiencies of the designs of section IX

A summary of the efficiencies of the various arrangements discussed in the last section, relative to that of ordinary randomised blocks, is given in Table XIV. The efficiencies in the limiting case in which the error variance is the same for all block sizes, *i.e.* when there is no association between neighbouring plots, are also given for comparison.

The pseudo-factorial arrangements have proved decidedly the most efficient in every case, more especially with 64 varieties. Even the $4 \times 4 \times 4$ arrangement, which was really only included to illustrate the computations, has been very successful, though for only 64 varieties and a small number of replications the 8×8 arrangements are to be preferred. Three-dimensional arrangements are likely to be most advantageous with a larger number of varieties, and in cases in which there are sufficient replications for the randomised blocks to be replaced by Latin squares.

It is not claimed that the uniformity trial on which the examples are based is necessarily typical of the soil heterogeneity ordinarily met with. It was in fact the first uniformity trial encountered in a search of the literature which contained the requisite number of plots of reasonable size. A certain number of trials were inspected which contained a large number of small plots ($\frac{1}{500}-\frac{1}{200}$ acre), and none of these exhibited such a large degree of soil heterogeneity. It may be doubted, however, whether trials of this nature fairly represent the type of land ordinarily available for experiment: in some cases, at least, it would seem that a specially uniform piece of land was chosen, often after inspection of the growing crop.

In any case the values of Table XIV for the case where there is no association between plots show that the pseudo-factorial arrangements

Table XIV. *Comparison of various types of arrangement.* (*Efficiencies expressed as a percentage of that of ordinary randomised blocks*)

	Chosen example	No association between plots
A. 49 varieties		
Randomised blocks of 49 plots	100	100
Blocks of 7 plots with 2 controls per block	84·0	53·2
Systematic controls every 5th or 6th plot (differences)	98·0	47·7
Systematic controls every 5th or 6th plot (covariance)	103·5	76·4
7 × 7 pseudo-factorial arrangement	126·4	80
7 × 7 Latin-square pseudo-factorial arrangement	133·1	84·2
B. 64 varieties		
Randomised blocks of 64 plots	100	100
Blocks of 8 plots with 2 controls per block	100·3	54·7
Systematic controls every 5th or 6th plot (differences)	111·9	47·7
Systematic controls every 5th or 6th plot (covariance)	118·0	76·4
4 × 4 × 4 pseudo-factorial arrangement	144·8	66·7
8 × 8 pseudo-factorial arrangement	150·0	81·8
8 × 8 Latin-square pseudo-factorial arrangement	157·2	85·7

are never likely to be much less efficient than systematic controls, and are always decidedly more efficient than random controls. They may be somewhat less efficient than ordinary randomised blocks when there is little soil heterogeneity to eliminate.

XI. Summary

A new method of arranging variety trials involving large numbers of varieties is described. This type of arrangement, for which the name *pseudo-factorial* arrangement is proposed, enables the block size to be kept small without the use of controls.

Various possible types of pseudo-factorial arrangement are discussed in detail and the necessary formulae developed. The appropriate methods of computation are illustrated by numerical examples based on the results of a uniformity trial on orange trees. It is shown that pseudo-factorial arrangements are likely to be more efficient than arrangements

involving the use of controls. In cases where there is considerable soil heterogeneity they are also markedly more efficient than randomised blocks containing all the varieties. In the chosen example gains in efficiency ranging from 26 to 57 per cent. were obtained.

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